HEALTHIER, LONGER, BETTER LIVES

Can OCR Rules Replace Azure OpenAI?

Medical receipt extraction benchmark

9 receipts · 6 fields · rules vs AI extraction

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Executive Summary

OCR + rules is not enough to replace AOAI across varied receipts.

96.1%

Azure DI + AOAI (prompt v2)

Development result

51.0–70.6%

OCR + multilingual rules

51.0% PaddleOCR Light · 70.6% Azure DI

Retrospective baseline

Why AOAI may still be needed

- Fewer rules to write and maintain for new layouts and scenarios
- Can potentially extract more fields for analytics (not yet benchmarked)
- Rules remain fine for simple, clearly labelled checks

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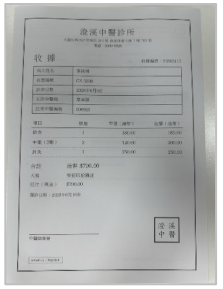
What We Compared

For each of the 9 receipts, we extract 6 key fields and compare the results across 3 approaches.

	1. Read the text (OCR)		2. Find the fields		
PaddleOCR Light + rules	**PaddleOCR Light** open source, own compute	→	**Hand-written rules**	→	**6 fields**
Azure DI + rules	**Azure Document Intelligence** cloud OCR	→	**Hand-written rules**	→	**6 fields**
Azure DI + AOAI	**Azure Document Intelligence** cloud OCR	→	**Azure OpenAI instructions**	→	**6 fields**


English


Bilingual


Chinese

9 receipts: 7 printed test receipts (English, bilingual, Chinese layouts) and 2 from the first benchmark round.

6 fields: doctor, registration no., patient, receipt no., service date, total.

51 of 54 possible fields were scored. 3 receipts do not contain a doctor registration number, so those fields were excluded.

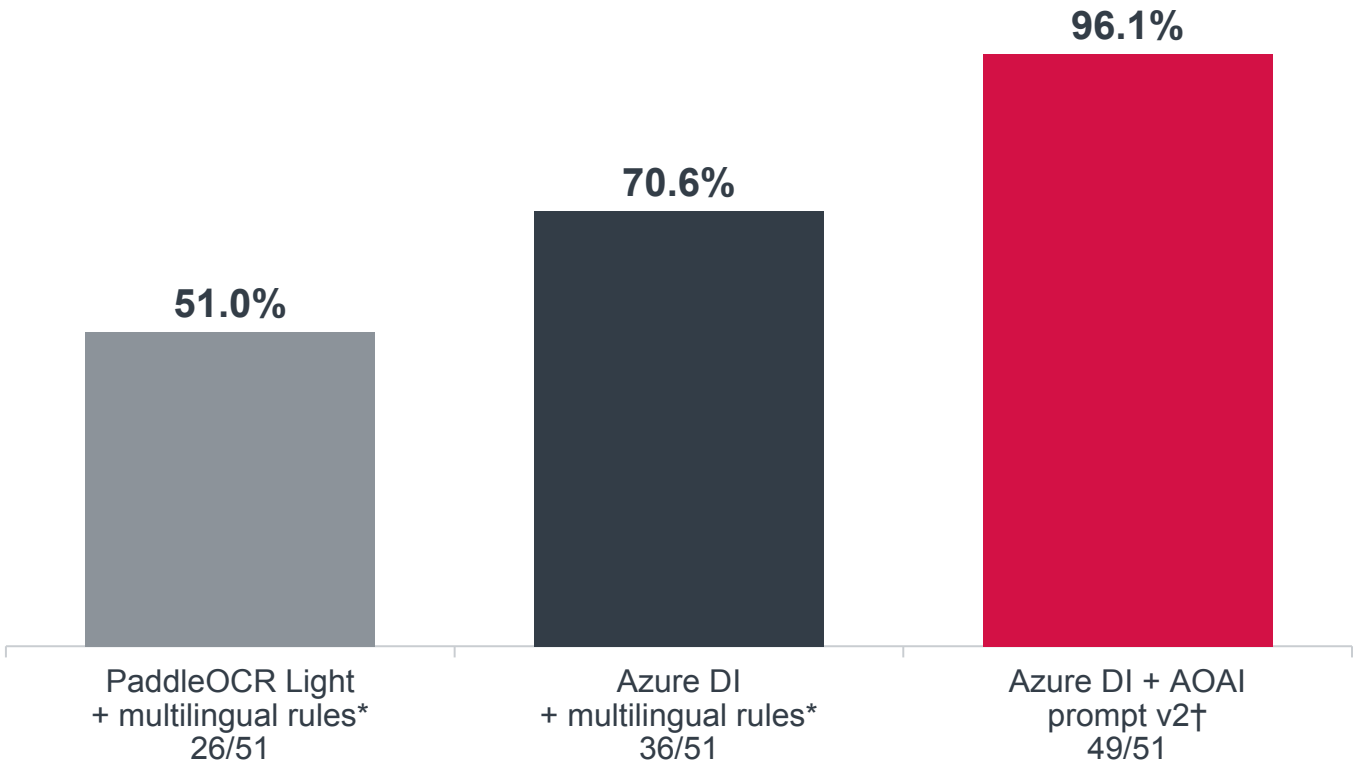
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Rules vs AOAI on the Same Receipts

MEASURED · DEVELOPMENT SET

Multilingual OCR rules reached 51.0–70.6%; Azure DI + AOAI reached 96.1%.



Multilingual rules*

- General rules with common Chinese labels
- No receipt-specific rules

To close the gap, rules would need

- New labels and layouts written by hand
- Retesting every receipt after each change

** Retrospective baseline † Development result (prompt refined on these receipts) Neither is independent validation. 9 receipts, 51 fields.*

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The Key Difference: Maintenance

MEASURED · DEVELOPMENT SET

Rules need us to anticipate every case in code. AOAI can often cope through plain-language instructions.

Manual extraction rules

A new label, language or layout appears → A developer anticipates it and writes rule code → Retest every receipt; a change can break a working one

Seen in this benchmark

- 69 → 225 label phrases written by hand
- 1 correct field broken by a rule change

AOAI extraction

New wording appears → Often handled by the existing instructions → If not: refine the instructions in plain language, rerun

Seen in this benchmark

- One prompt revision: 4 fields fixed, 0 broken

Where rules were enough: a yes/no check for a diagnosis comment under clear labels. Rules matched AOAI on 13/13 receipts (development result).

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Advantages and Disadvantages

Rules win on cost, auditability and predictability; AOAI on accuracy, maintenance and flexibility.

	OCR + rules	Azure DI + AOAI	Evidence
Cost	+ PaddleOCR: own compute, no API fee (Azure DI: page fee)	− Azure DI page fee + ~962 input / 243 output tokens per receipt	*Prices to confirm*
Speed	PaddleOCR 13.9 s (CPU) · Azure DI 3.8 s	6.4 s per receipt	Measured
Accuracy	− 51.0–70.6% with multilingual rules (retrospective)	+ 96.1% with prompt v2 (development result)	Measured
Maintenance	− New labels, languages and layouts need code and a full retest	+ Many changes made in plain-language instructions; still retest	*Measured / expected*
Flexibility	− Finds only what it was written for	+ New fields can be requested in the prompt (not yet benchmarked)	*Expected*
Auditability	+ Every result traceable to a rule	− Harder to explain; keep quoted evidence and logs	*Expected*
Reliability	+ Same input, same output	− Can return a confident wrong value (e.g. a sub total as the total); model updates need retesting	*Measured / expected*

Green = stronger side. Multilingual rules include common Chinese labels (retrospective baseline). Expected = not measured here.

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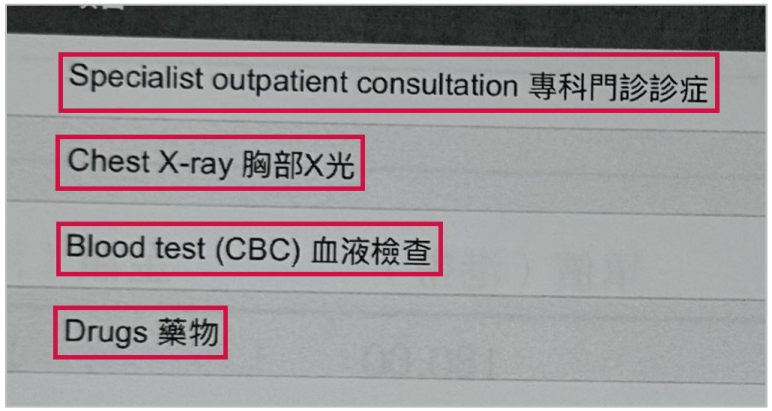
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Beyond Six Fields: From Receipts to Insights

POTENTIAL · NOT YET TESTED

Today we extract 6 fields. AOAI could also capture what was treated, tested and prescribed.

1. What the receipt shows



2. What AOAI could extract

Item	Type	HK$
Specialist consultation	**Consultation**	950.00
Chest X-ray	**Test**	620.00
Blood test (CBC)	**Test**	380.00
Drugs	**Medication**	245.50

Illustration typed from the receipt text, not AOAI output

3. What we could analyse

Cost by treatment
Spend on consultations, tests and medicines

Provider comparison
Typical charges per test across providers

Trends over time
Which treatments and medicines are rising

Unusual claims
Combinations or amounts that stand out

Potential, not yet tested. Only the 6 current fields have been benchmarked. Diagnoses, symptoms, treatments and medications would need a pilot.

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Recommendation

Use rules for simple, labelled checks. Use AOAI where receipts vary or richer data is needed.

Simple OCR rules are sufficient when…

- Layouts are few, known and stable
- Fields have clear printed labels
- The question is simple, e.g. is a diagnosis comment present?
- No API fee and fully traceable output are required

AOAI adds real value when…

- Providers, layouts and languages vary (Chinese, bilingual)
- Fields lack consistent labels: names with titles, totals vs subtotals
- We want less scenario-specific code to maintain
- Richer medical data is wanted (potential, not yet benchmarked)

1. Freeze

the rules and AOAI prompt v2

2. Validate

30+ unseen, real, anonymised receipts; count rule changes

3. Pilot extra fields

medications, treatments, symptoms, diagnoses with ground truth

4. Confirm cost

Document Intelligence pages and gpt-5-mini tokens

Evidence so far: 9 receipts; results are development or retrospective, not yet validated on unseen receipts.

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